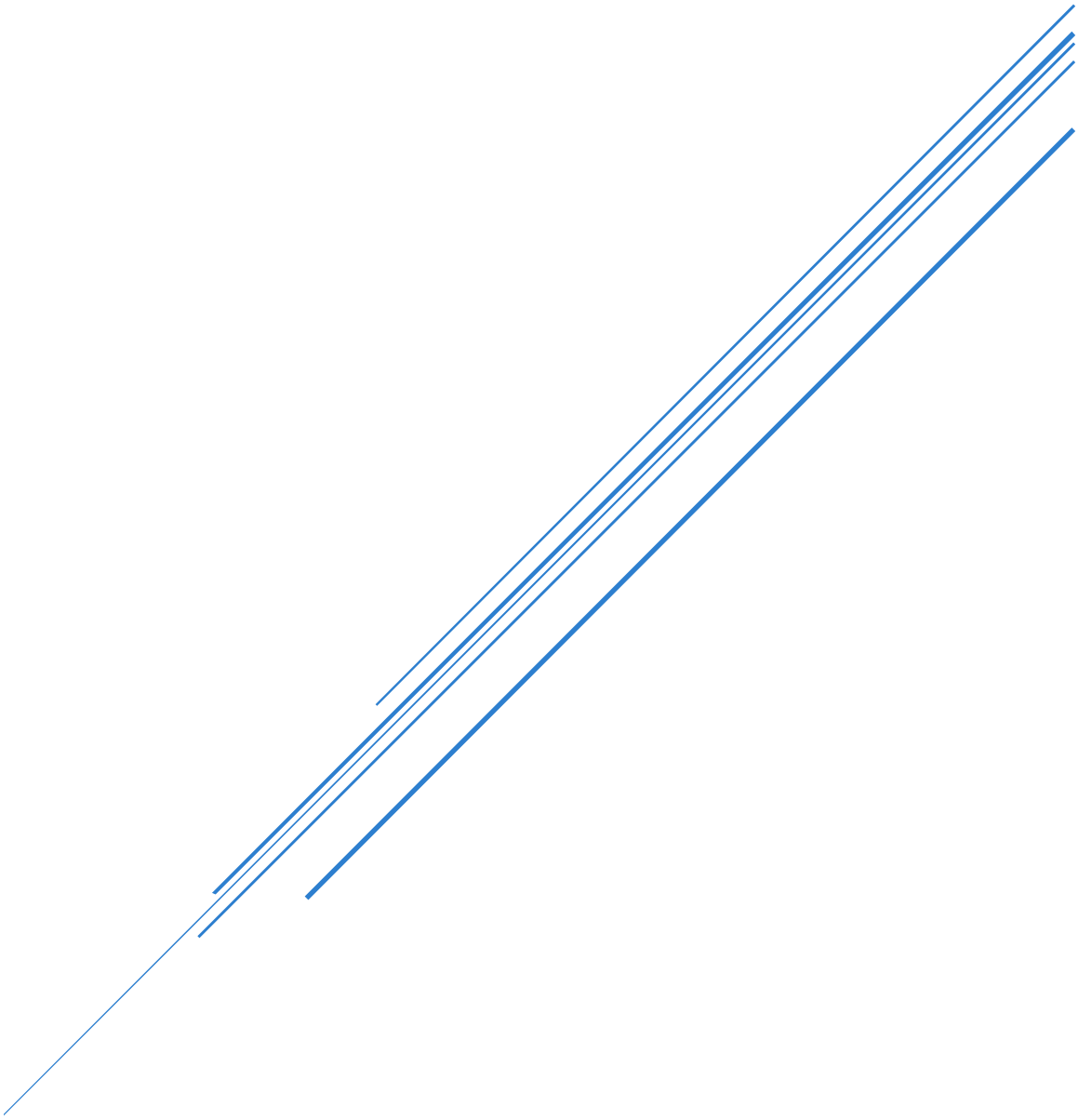
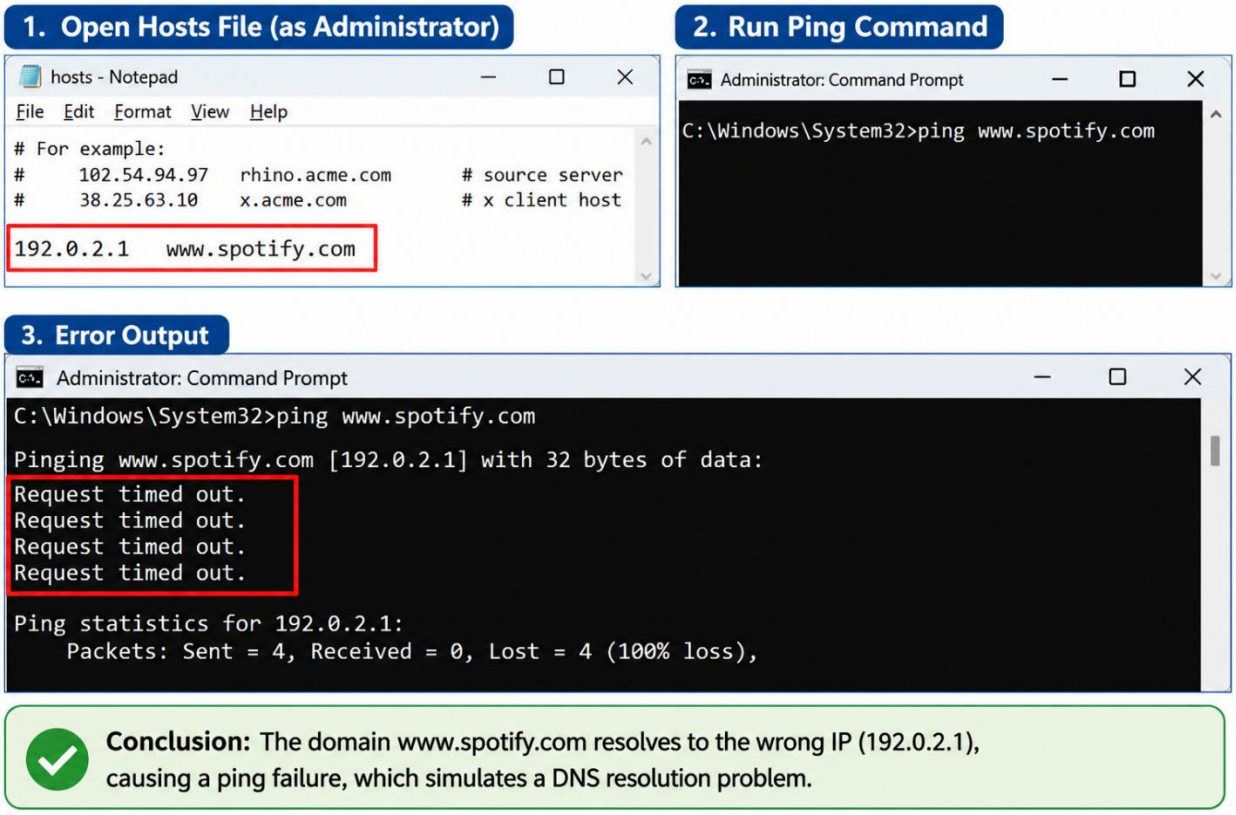
DAY - 5 TROUBLESHOOTING NETWORK SERVICES

Assignment - 13



1. **Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.**

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The DNS resolution failure was simulated by adding an incorrect IP address (192.0.2.1) for www.spotify.com in the Windows hosts file. After running the command ping www.spotify.com, the domain resolved to the wrong IP address.

Error Message:

Pinging www.spotify.com [192.0.2.1] with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Result: The ping failed because the domain was mapped to an incorrect IP address.

2. **Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.**

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1. Network Configuration:

A DHCP server was configured in Cisco Packet Tracer to assign IP addresses automatically to client devices. An incorrect default gateway (192.168.1.254) was intentionally configured in the DHCP pool instead of the correct gateway (192.168.1.1).

2. Client Verification:

The client PC was connected to the network and configured to obtain an IP address automatically using DHCP. The ipconfig command was used to check the assigned network settings.

Example Output:

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.254

3. How to Identify the Issue:

- **Run ipconfig on the client PC.**
- **Check whether the default gateway is correct.**
- **Use ping 192.168.1.1 to test connectivity to the actual gateway.**
- **Check the DHCP pool configuration on the server.**
- **Verify that the router's LAN IP address is 192.168.1.1.**

4. How to Fix the Issue:

- **Open the DHCP server configuration.**
- **Change the incorrect default gateway from 192.168.1.254 to 192.168.1.1.**
- **Renew the client's IP configuration using ipconfig /release and ipconfig /renew.**
- **Run ipconfig again to verify the corrected gateway.**

Result: The client initially received an incorrect default gateway from the DHCP server. After correcting the DHCP configuration and renewing the IP address, the client received the correct gateway.

3. **Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.**

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1. Troubleshooting Steps:

1. **Open Command Prompt on the client PC.**
2. **Run ipconfig to check the IP address and DNS server address.**
3. **Run nslookup www.google.com to test DNS resolution.**
4. **If the DNS server is incorrect or not responding, check the DNS server configuration.**
5. **Verify that the DNS service is running and the DNS server has the correct records.**
6. **Check network connectivity between the client and DNS server using the ping command.**

2. Example NSLOOKUP Output:

C:\> nslookup www.google.com

DNS request timed out.

Server: Unknown

Address: 192.168.1.254

DNS request timed out.

This indicates that the configured DNS server is not responding.

3. Solution:

- **Configure the correct DNS server IP address in the DHCP settings.**
- **Make sure the DNS service is running.**
- **Verify the DNS records.**

- **Renew the client's IP configuration using ipconfig /release and ipconfig /renew.**
- **Test again using nslookup www.google.com.**

Result: After correcting the DNS server address and configuration, the client can successfully resolve domain names into IP addresses.

4. **A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.**

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1. Possible Causes:

1. **DHCP server is switched off or unavailable.**
2. **DHCP scope or IP address pool is exhausted.**
3. **Network cable is disconnected or the switch port is down.**
4. **Incorrect DHCP configuration or IP address range.**
5. **DHCP service is disabled.**
6. **Incorrect VLAN configuration or missing DHCP relay (ip helper-address).**

2. Step-by-Step Troubleshooting Process:

1. **Check Network Connectivity: Verify that the network cable is connected and the switch port is active.**
2. **Check IP Configuration: Run ipconfig to confirm that the device has an APIPA address.**
3. **Check DHCP Server: Make sure the DHCP server is powered on and its DHCP service is running.**
4. **Check DHCP Scope: Verify that the DHCP pool has enough available IP addresses.**
5. **Check DHCP Configuration: Verify the IP address range, subnet mask, and default gateway.**
6. **Check VLAN and Relay Configuration: Ensure the client is in the correct VLAN and DHCP relay is configured if the server is on another network.**
7. **Renew IP Address: Run the following commands:**

ipconfig /release

ipconfig /renew

8. Verify the Result: Run ipconfig again to check whether the device has received a valid IP address.

3. Example of Correct IP Configuration:

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DHCP Server: 192.168.1.2

Result: After fixing the DHCP server or network configuration, the client receives a valid IP address instead of an APIPA address.

5. **Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.**

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1. Common DNS Issues

DNS Issue	Symptoms	Possible Causes	Recommended Solutions
DNS Server Not Responding	Websites do not open by domain name.	DNS server is offline or unavailable.	Check the DNS server and restart the DNS service.
Incorrect DNS Records	Website redirects to the wrong IP address.	Incorrect A record or hosts file entry.	Correct the DNS records or hosts file entry.
DNS Resolution Failure	Websites open using IP addresses but not domain names.	Incorrect DNS server IP or DNS configuration.	Configure the correct DNS server and test using nslookup.

2. Common DHCP Issues

DHCP Issue	Symptoms	Possible Causes	Recommended Solutions
APIPA Address	Device receives an IP address starting with 169.254.x.x.	DHCP server is unavailable or not responding.	Check network connectivity and restart the DHCP service.
DHCP Scope Exhaustion	New devices cannot obtain an IP address.	All available IP addresses in the DHCP pool are assigned.	Increase the DHCP pool range or remove unused leases.
Incorrect Default Gateway	Device receives an IP but cannot access other networks or the internet.	Wrong default gateway configured in the DHCP pool.	Correct the default gateway in DHCP settings and renew the IP address.